Www.Medicalstudyzone.com

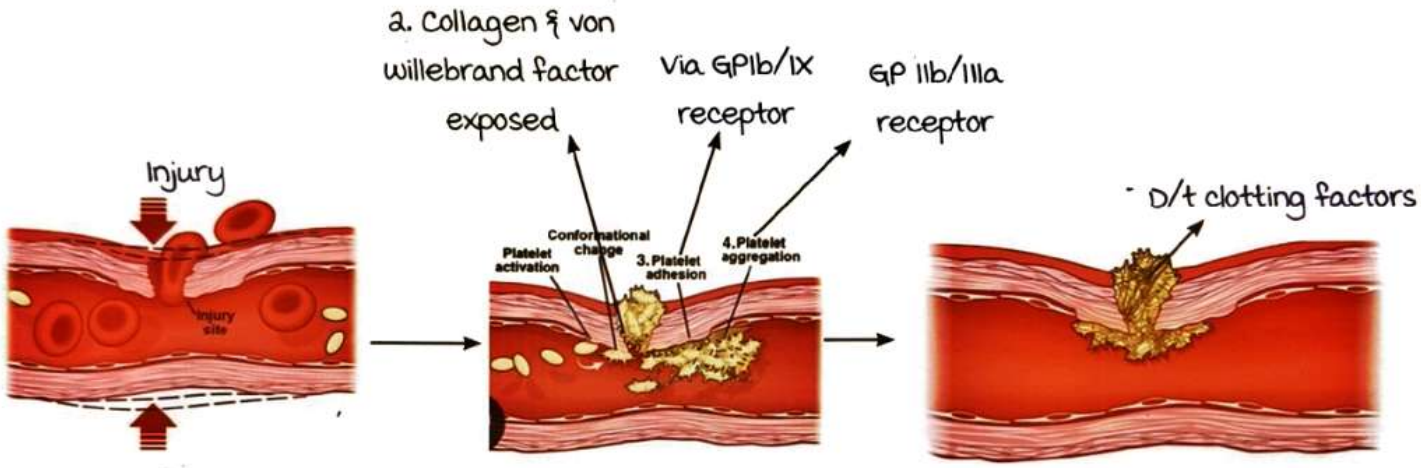
----- Active space -----

VASCULAR INJURY

Coagulation

00:01:15

HEMOSTASIS :



1. Vasoconstriction :

Temporary stop in blood flow.

Mediators :

- Serotonin (From platelets)
- Endothelin

Primary hemostatic plug

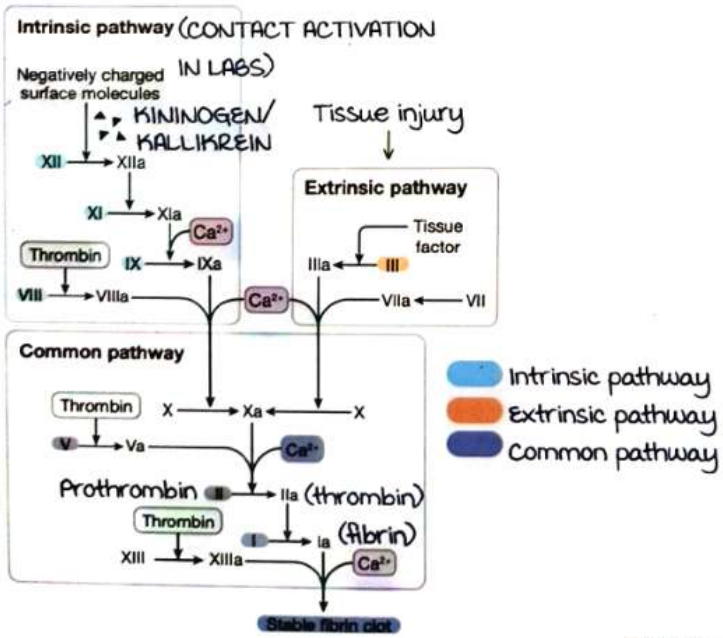
Definitive clot

Clot formation

Requires: 1. Platelets.
2. Clotting factors.
3. Vascular endothelium.

COAGULATION CASCADE :

Positive feedback mechanism..



Clotting factors :

I	Fibrinogen
II	Prothrombin
III	Tissue thromboplastin
IV	Ca^{+a}
V	Proaccelerin
VII	Proconvertin
VIII	Anti-hemophilic factor A
IX	Anti-hemophilic factor B (Christmas factor)
X	Stuart-Prower factor
XI	Anti-hemophilic factor C
XII	Hageman factor
XIII	Fibrin stabilizing factor

Www.Medicalstudyzone.com



Applied aspect :

----- Active space -----

Clotting factors	Disease in deficiency
Fibrinogen	Afibrinogenemia
Prothrombin	Hypoprothrombinemia (2° to Vitamin K deficiency)
Proaccelerin	Parahemophilia
Proconvertin	Hypoconvertinemia
Anti-hemophilic factor A	Hemophilia
Anti-hemophilic factor B	Christmas disease
Anti-hemophilic factor C	Plasma thromboplastin antecedent (PTA) deficiency
Hageman factor	Hageman triad

COAGULATION PROFILE :

Activated partial thromboplastin time (APTT) vs Prothrombin time (PT) :

	APTT	PT
Substances used	1. Kaolin : XII → XIIa. 2. Cephalin : Substitute for platelet phospholipids. 3. Ca^{+2}.	1. Exogenous tissue thromboplastin. 2. Ca^{+2}.
Pathway assessed	Intrinsic & common pathway	Extrinsic & common pathway
Normal range	26-40 S	11-16 S
Defect/deficiency of factors	VIII, IX, XI, XII, V, X Fibrinogen & prothrombin	VII, V, X, fibrinogen & prothrombin

3. Platelet count : 1.5 – 4 lakh cells/mm^3.
4. Bleeding time : 2-8 mins.
5. Clotting time : 3-6 mins.

Anticoagulation

00:23:50

ANTICOAGULATION PATHWAYS :

mechanism :

1. Thrombin + thrombomodulin :

⊕ | Activates ⊕

- Protein C.
- Protein S.
  - ↓ Inactivates
  - Factors: V, VIII

tPA (Tissue plasminogen activator)
↓
Plasminogen → Plasmin/fibrinolysin

2. Antithrombin III + heparin:
   ↓ Inactivates
   Factors : IX, X, XI, XII

3. Tissue factor pathway inhibitor (TPI).

Note : Thrombomodulin is present in cerebral circulation.

Feedback

Www.Medicalstudyzone.com



----- Active space -----

Factors regulating anticoagulation pathways :

1. Plasminogen activator inhibitor.
2. α2-antiplasmin.
3. Thrombin activatable fibrinolysis inhibitor.

Note : Role of vitamin K in coagulation.

Warfarin ⊖

Glutamic acid —Vitamin K→ γ carboxyglutamic acid —Activates→ Factors : II, VII, IX, X
Protein C & protein S.

Feedback